

Inequalities and Equations with Decimals (Homework)

Solve. You may use a calculator. Round to the nearest thousandth if necessary.

① $2.3x - 4 = -1.8x + 2.3$

$x \approx 1.537$

② $\frac{y}{1.3} - 6 = -4.7$

$y = 1.69$

③ $-7.7z + 2.5z + 1 = 3.8 - 1.6z$

$z \approx -0.778$

④ $6.6 = \frac{x}{-2} + 5.9$

$x = -1.4$

⑤ $4 - \frac{y}{0.5} = -3.68 - 2.1$

$y = 4.89$

⑥ $0.004z - 0.01z - 6 = 2.03z$

$z \approx -2.947$

⑦ $2.2(4.1x + 3) = -2(-3x + 0.9)$

$x \approx -2.781$

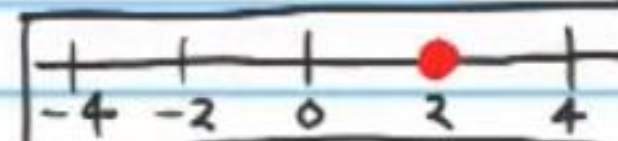
⑧ $7.5 - (3 - 4x) = 2 - 1.1(4.6x + 2.7)$

$x \approx -0.604$

Find the algebraic solutions and graphical solutions of the following equations.

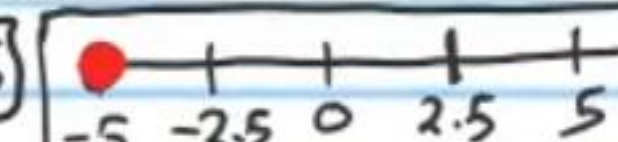
⑨ $1 = 4x - 7$

$x = 2$



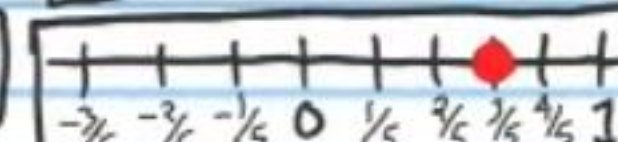
⑩ $-5z + 4z = 7 - 2$

$z = -5$



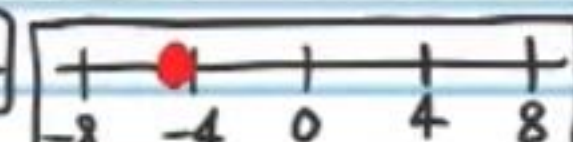
⑪ $-3 = -5y$

$y = \frac{3}{5}$



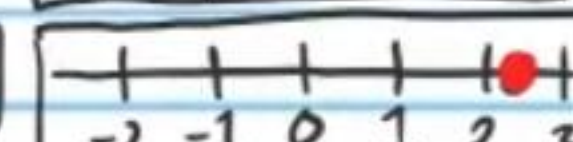
⑫ $1.8 = \frac{x}{-2} - 0.4$

$x = -4.4$



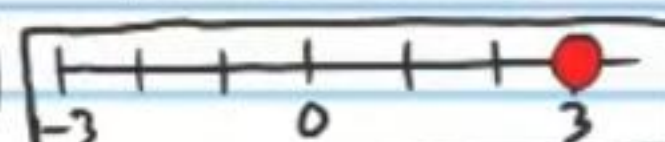
⑬ $5 - \frac{y}{0.6} = 1$

$y = 2.4$



⑭ $4 = -2x - 2x + 16$

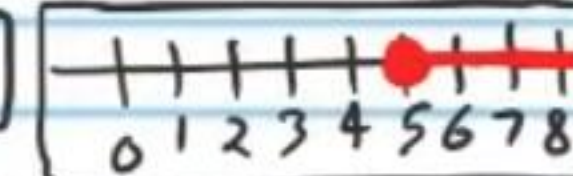
$x = 3$



Find the algebraic solutions and graphical solutions of the following inequalities.

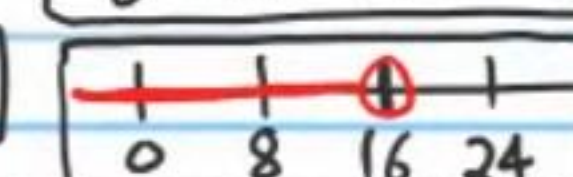
⑮ $-2 + 5x \geq 13 + 2x$

$x \geq 5$



⑯ $-3 + 1 < \frac{y}{-4} + 2$

$16 > y$



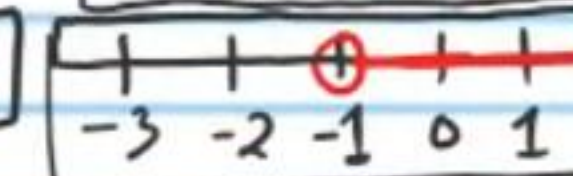
⑰ $3(2x - 4) \leq 1 - 8$

$x \leq \frac{5}{6}$



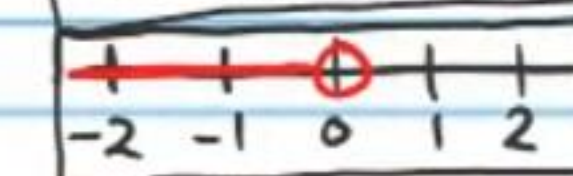
⑱ $-7z - 3z + 1 < -8z + 3$

$z > -1$



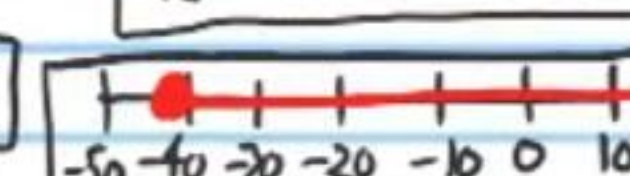
⑲ $-3x + x - 7 + 5x < 2x - 7$

$x < 0$



⑳ $4 - 8 + 1 \leq \frac{z}{3} + 11$

$-42 \leq z$



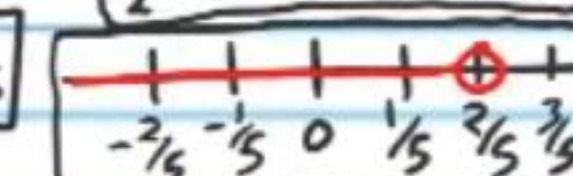
㉑ $6x - 9x - 2x - 4 \leq 2 - 6x + x + 4x$

$-\frac{3}{2} \leq x$



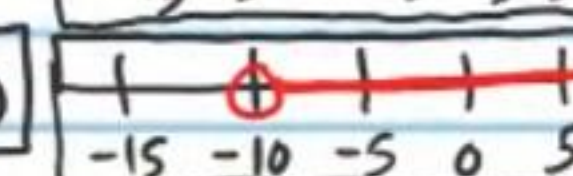
㉒ $3 + 5y - 8y > 2y + 1$

$y < \frac{2}{5}$



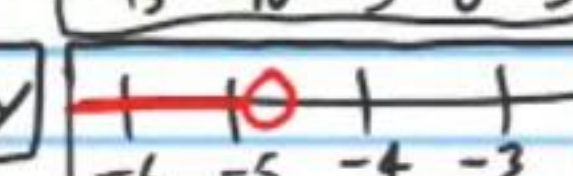
㉓ $\frac{x}{-5} + 2 < 4$

$x > -10$



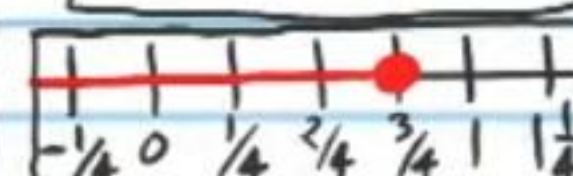
㉔ $-1 > 3 + \frac{y}{1.2}$

$-4.8 > y$



㉕ $6 + 2(z - 3) \geq 3(2z - 1)$

$z \leq \frac{3}{4}$



㉖ $-3 + 4(-2x - 2) \leq 1 + (-5)$

$x \geq -\frac{7}{8}$



㉗ $-(3 - 2y) < 4y - 9y$

$\frac{3}{7} > y$

